FIREWORKS × NVIDIA · NOVEMBER 2026

Emberz

A guardian on Grandma’s phone line.

A fine-tuned model listens during the call, warns the person being targeted, and calls the family before the scammer gets the code.

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THE PROBLEM

The scam happens on the call. Nothing listens there.

$7.7B

lost by Americans 60+ in 2025. **37% of all fraud losses**, from 20% of complaints.

3×

bigger loss when AI is involved: **$112K** per complaint for victims 60+, vs $38K.

$0.34

is what **our own AI scam call** costs: an LLM script, a stock voice, a phone API.

FBI IC3 2025 Annual Report. Caller-ID filters judge the number before it rings; the damage happens during the conversation.

EXAMPLE: A REAL CALL, 2 NOV 2026

An AI scammer calls “Margaret”. 94 seconds.

0:30	“This is Daniel from the Fraud Prevention Team at First Meridian.”	**AI voice detected** (score 0.9995)
0:58	“We’ve flagged a charge of eight hundred forty two dollars… Did you make this purchase?”	**Urgent: fake bank call.** Only Margaret hears: “Don’t confirm anything. Hang up and call the number on your card.” Family phoned.
1:23	“I’m sending a verification code to your phone right now.”	**Urgent: code request.** The caretaker, listening in since 1:16, hears it.
1:28	Margaret: “Okay, let me see.”	**Caretaker presses 9.** Scammer dropped, call over. No code shared.

AI caller over SignalWire → our Telnyx guarded number → a real phone. Quotes from the live transcript; times from the call log (recordings/, results/ledger-numan.md).

HOW IT WORKS

Hear it, judge every turn, act.

Guarded number
Telnyx bridges the call to the real phone

→

Hears it
NVIDIA Nemotron Speech, live, on a laptop

→

Judges each turn
Our guardian, fine-tuned on Fireworks

→

Acts
Whisper, family call (1 listen, 0 talk, 9 end)

What the guardian returned at 0:58

```
risk: "urgent"
category: "scam.impersonation_bank_gov"
coach_line: "Don’t confirm anything. Hang up and call the number on your card."
caretaker_note: "Caller claims a suspicious charge. This is a common scam tactic."
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It reads the **family’s own rules** (“alert me about any money talk, even with family”), the last 8 turns and the AI-voice score.

Trained on **1,440 generated calls**, about 28% of them harmless look-alikes that reuse scam words.

THE BENCHMARK

More calls handled right than Kimi K3.

90.3%

Emberz guardian

83.7%

Kimi K3, frontier

p = 0.003 (McNemar). Every gain over every baseline holds, p ≤ 0.005.

4.5 turns: median warning before the caller asks for the code or money.

257 held-out calls by NVIDIA Nemotron 3 Ultra, not our training writer. Handled = scam caught before the ask, or harmless call left alone.

WHERE IT WINS

Quieter, and it follows the family's rules.

False alarms

18%

Kimi K3 23%. The same model untuned: 53%.

Family rules followed

19/22

Kimi K3 4/22. No baseline gets more than 5.

Public data, never seen

98/100

scams caught on BothBosu, with 0/100 false alarms.

RULE-FLIP EXAMPLE

Grandson asks to borrow rent money. With the rule **"alert me about any money talk, even with family"** → alert. Without it → stay quiet. Same words, different family, different answer.

All 257 held-out calls (79 harmless, 22 rule-flip pairs). Wilson 95% CIs in the paper: v2 false alarms 11–28%, Kimi K3 15–33%.

WHY IT'S WORTH IT

Small cost per family. Big pool of losses.

$38.6M

a year saved for **every 1%** of call-driven scams stopped (tech support, government impostors, romance: $3.86B a year).

$3.64

per household per month to run the line and the model (modeled). **One stopped $3,000 loss covers 25 years** at $10 a month.

The buyer: adult children of an aging parent. Phone-maker and carrier tools warn only the person being scammed, or stop at the ring. Emberz brings in the family.

Losses: FBI IC3 2025. Running cost: docs/business_value.md (Telnyx list prices, 100 call-minutes per household-month).

HONEST LIMITS

What we haven’t solved yet.

Serving drift

The dedicated deployment follows family rules less well than the checkpoint we tested. Raised with Fireworks.

Synthetic test calls

The benchmark is generated. Real role-played calls are the next test set.

Not for robocalls

One-way recorded pitches have no conversation to read; carriers already block them.

Consent built in: every call starts with a recorded notice, the protected adult opts in themselves, no voiceprints, deleted after 7 days.

EMBERZ

Give Grandma a guarded number.

90.3% of calls handled vs 83.7% for Kimi K3. Fewer false alarms. Follows the family's rules. Working on real phones today.

NVIDIA Nemotron hears the call · Fireworks trains and serves the guardian

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github.com/matt-ceran/emberz